
Agency MBS Fixed-rate Prepayment Model: Technical Documentation

Cello Research

May 11, 2018

1 Introduction

This document provides a focused technical discussion of the Agency fixed-rate prepayment model developed by Cello Research. The contents are organized as follows. Section 2 describes the data set on which the model is estimated, Section 3 discusses the specification of the model, and Section 4 concludes by discussing how the model is calibrated. The Appendices collect useful background material. Further detail on the underlying drivers of prepayment behavior and the selection of explanatory variables used to model these drivers can be found in Chapter 5 of [Cello16].

The model structure described below is for Fannie Mae and Freddie Mac 30-year pools but should carry over with little to no modification for other product types: Ginnie Mae 30-year, Fannie Mae 15-year, Hybrid ARMs etc. The overarching goal is for differences in the prepayment behavior of different mortgage products to be captured mostly through different parameter values produced in our estimation process rather than through an alternative specification.

2 Data Set

2.1 Loan- and Pool-Level Data

There are two fundamental data sets that can be used to estimate the prepayment model: **pool-level (PL)** and **loan-level (LL)** data. While LL data set provides more granularity, it spans a shorter history,¹ and the opposite holds for PL data. Consequently, this difference presents us with a fundamental tension in deciding which data set to pick in fitting the model: Should we trade-off superior granularity for shorter history? A shorter time period means that we sample a much more limited range of macroeconomic environments (interest rates, mortgage credit availability etc.) when calibrating the model. Thus, whenever we encounter a “new” environment, model projections presumably are much less accurate than if the fit originally incorporated instances of this environment. On the other hand, given that prepayment behavior does not remain constant over time, more recent data potentially provide a better indication of the current prepayment response curve and will provide a better “local” fit.

There is no panacea for these inconsistencies. Prepayment modelers rely on their domain knowledge to make these judgment calls and we discuss the rationale behind our baseline fitting data set:

¹Freddie Mac loan level data is available from September 2006 while Freddie Mac pool level data is available from the mid-80s (although only since October 1994 in our database). See Appendix 5 for a detailed description of the coverage of Fannie and Freddie Mac data. Because much more history is available in Freddie Mac LL data, LL fits tend to rely on this data set.

- **Use Loan-level replines** (essentially modeling prepayment data from September 2006 onwards): This allows for better modeling of dispersion and 10+ years of prepayment history means that a fairly comprehensive set of economic environments is in sample including a brief snapshot of the pre-Housing crisis period.
- **Only FGLMC tickers** (as opposed to simultaneously attempting to fit FGLMC, CQ/CR and Jumbo): The other sectors will be layered on eventually but the overriding objective is to start by getting the “generics” right.
- **Pre-2010, use all available LL data; post-2010, use only mortgages originated after December 2009 (that also meet the ticker constraint)**: The refinance response of HARP-eligible loans is special in some sense because there was a targeted government program to refinance such loans. With the eventual sunset of the program and the ever-decreasing pool of eligible borrowers, it makes sense to exclude these loans from the fit. In particular, existing fits are finding a middle-ground between post-HARP and HARP-eligible collateral that does a poor job of capturing the true behavior of either subset of borrowers. On the other hand, it is useful to include some pre-HARP data so that the existing fits incorporate at least some sense of what the pre-crisis refinance response was.

This is not a complete characterization of our fitting data set which in practice often uses a combination of LL and PL data; Section 4 has further details.

2.2 Replines

Technically speaking, we do not directly use LL or PL data for fitting the model but **replines**. Given a collection of loans backing an MBS, **stratification** is the process of dividing members of this collection into homogeneous and mutually exclusive subgroups. Each subgroup can then be appropriately modeled as a single “average” loan called a **repline** that can be considered a representative of this subgroup. The idea behind repline creation is the reduction in dimensionality. For example, 1 million loans can be reduced to 1,000 replines which are much more tractable from a computational perspective when predicting prepayments.

Replines can be implemented in several different ways. For example, in our current production system, we have: (1) **Single replines** that correspond to a simple loan-level weighted average; (2) **Pool-level replines** that stratify an MBS into a number of subgroups based on pool-level data; and (3) **Loan-level replines** that do the same as (2) but use loan-level data. In effect, we get increasing granularity but also increasing use of computational resources as we move from (1) to (3).

2.3 Auxiliary Data

The LL and PL data sets provided by the GSEs are the starting point for prepayment analysis and are enriched in two ways. First, existing mortgage and borrower attributes are combined and/or integrated with macroeconomic data to generate new loan-/pool-specific variables such as **burnout**, **CLTV**, **incentive**, **pct_HARPed**, and **refinancingcost** among others. In addition, factor dates are merged with macroeconomic time series such as **HPA Momentum**, **Media Effect**, and **Mortgage Credit Availability**. Appendix 6 contains further details.

3 Model Specification

Given a collection of borrower/loan characteristics associated with a repline, **repline**, and a macroeconomic environment (typically expressed through a vector of forward projections for mortgage rates, credit availability, and house prices), **macroenv**, a prepayment model generates a vector of prepayment rates,

$[\text{Prepay}(\text{repline}, \text{macroenv})]_{t=0}^{\text{Term}}$, starting from the origination date of the repline ($t = 0$) and ending at its maturity ($t = \text{Term}$). In our model, we specify **Prepay** as the sum of five different (mutually exclusive) sources of prepayment and develop explicit submodels for each of these different sources:

1. Curtailments and Full Payoffs (denoted as **Curt**);
2. Defaults (**Dflt**);
3. Out-of-the-money Cash-out Refinancings (**Cout**);
4. Turnover (**Turn**); and,
5. Refinancings (**Refi**).

In equations, the projected prepayment rate **Prepay** is given as:

$$\text{Prepay} = \text{Curt} + \text{Dflt} + \text{Cout} + \text{Turn} + \text{Refi}. \quad (1)$$

The next step is to describe the specification of each of these submodels and the explanatory factors included in them, which we do below. One of the key features of our model is the use of spline functions² to capture the complex interaction between our explanatory variables and the response function (prepayment rates). Each “Effect” in the tables below corresponds to a spline function unless otherwise indicated. The explanatory factors themselves can sometimes be transformations of other more “primitive” variables.

3.1 Curtailments and Full Payoffs

$$\text{Curt} = \text{Curt_WALA} \times \text{Curt_CLTV} \times \text{Curt_FICO} \quad (2)$$

Effect	Function of	Comments
Curt_WALA	WALA	Seasoning
Curt_CLTV	CLTV	Current LTV
Curt_FICO	FICO	Credit Score

Table 1: Specification of the Curtailment and Full Payoff Submodel

3.2 Defaults

$$\text{Dflt} = \text{pct_dq} \times \text{Dflt_Delq} + (1 - \text{pct_dq}) \times \text{Dflt_Curr} \quad (3)$$

where **Dflt_Delq** and **Dflt_Curr** are submodels for the population of delinquent³ and current loans respectively. The Default Delinquency submodel is defined as:

$$\begin{aligned} \text{DfltDelq} = & \text{DfltDelq_Base} \times \text{DfltDelq_CAI} \times \text{DfltDelq_WALA} \times \text{DfltDelq_monthsSince} \\ & \times \text{DfltCurr_CLTV} \times \text{DfltCurr_FICO} \end{aligned} \quad (4)$$

The Default Current submodel is defined as:

²As opposed to a “general” functional form such as the general logistic function (GLF) that can accomodate a large number of shapes through an appropriate choice of its parameters. The tradeoff here is between emphasizing local accuracy through choosing spline knots but then potentially generating a curve that has kinks and wobbles as opposed to fitting a smoother curve that is not as accurate. We attempt to deal with this tension by sometimes post-processing spline knot points through a GLF to generate a new set of knot points.

³Defined here as the sum of 30 Day, 60 Day, 90 Day, and 120 Day delinquent pool percentages.

Effect	Function of	Comments
DfltDelq_Base	NA	Baseline Constant
DfltDelq_CAI	CAI	Mortgage Credit Availability Index
DfltDelq_WALA	WALA	Seasoning
DfltDelq_monthsSince	monthsSince	Months since last recorded Delinquent status
DfltDelq_CLTV	CLTV	Current LTV
DfltDelq_FICO	FICO	Credit Score

Table 2: Specification of the Default Delinquency Submodel

$$\begin{aligned} \text{DfltCurr} = & \text{DfltCurr_Base} \times \text{DfltCurr_CAI} \times \text{DfltCurr_WALA} \times \text{DfltCurr_monthsSince} \\ & \times \text{DfltCurr_CLTV} \times \text{DfltCurr_FICO} \end{aligned} \quad (5)$$

Effect	Function of	Comments
DfltCurr_Base	NA	Baseline Constant
DfltCurr_CAI	CAI	Mortgage Credit Availability Index
DfltCurr_WALA	WALA	Seasoning
DfltCurr_monthsSince	monthsSince	Months since last recorded Current status
DfltCurr_CLTV	CLTV	Current LTV
DfltCurr_FICO	FICO	Credit Score

Table 3: Specification of the Default Current Submodel

3.3 Turnover

$$\begin{aligned} \text{Turn} = & \text{Turn_Base} \times \text{Turn_CAI} \times \text{Turn_HPA2yr} \times \text{Turn_WALA} \\ & \times \text{Turn_Lockin} \times \text{Turn_WACLS} \times \text{Turn_FICO} \times \text{Turn_pct_REFI} \\ & \times \text{Turn_Tax} \times \text{Turn_SATO} \times \text{Turn_Seasonals} \end{aligned} \quad (6)$$

Effect	Function of	Comments
Turn_Base	NA	Baseline Constant
Turn_CAI	CAI	Mortgage Credit Availability Index
Turn_HPA2yr	HPA	House price appreciation momentum
Turn_WALA	WALA	Seasoning
Turn_Lockin	Incentive	Economic disincentive
Turn_WACLS	WACLS	Current Loan Size
Turn_FICO	FICO	Credit Score
Turn_pct_REFI	%Refinance	Loan Purpose
Turn_Tax	Calendar time	Tax multiplier
Turn_SATO	SATO	Spread-at-origination
Turn_Seasonals	Seasonals	Seasonal factors for turnover

Table 4: Specification of the Turnover Submodel

3.4 Refinancings

There are two S-curves: one for a “low” Media Effect (ME) environment ($\%Refinanceable < 30$) and one for a “high” environment ($\%Refinanceable > 84$). The Media Effect is interpreted as a multiplier between 0 and 1 that gives us an S-curve that is a blend of the “low” and “high” S-curves. In other words, for a given incentive, the aggregate refinancing S-curve is given by $(1 - ME) \times S_{lowmedia} + ME \times S_{himedia}$.

$$\begin{aligned} Refi = & Refi_Base \times Refi_CAI \times Refi_HPA2yr \times Refi_Burnout \\ & \times Refi_WACLS \times Refi_refi_elig_pct \times Refi_pct_tpo \times Refi_pct_HARPed \\ & \times Refi_HARP_eligible \times Refi_WALA \\ & \times [(1 - ME) \times Refi_LoMedia + ME \times Refi_HiMedia] \end{aligned} \quad (7)$$

Effect	Function of	Comments
Refi_Base	NA	Baseline Constant
Refi_CAI	CAI	Mortgage Credit Availability Index
Refi_HPA2yr	HPA	House price appreciation momentum
Refi_Burnout	Burnout	Accumulated positive incentive
Refi_WACLS	WACLS	Current Loan Size
Refi_refi_elig_pct	%RefiEligible	Eligible to refinance
Refi_pct_tpo	%TPO	Origination Channel
Refi_pct_HARPed	%HARPed	How many loans went through HARP
Refi_HARP_eligible	%HARP Eligible	How many loans are HARP eligible
Refi_WALA	WALA	Seasoning
Refi_SCurve	Costs, Incentive, Media Effect	S-Curve for refinancing
Refi_Seasonals	Seasonals	Seasonal factors for refinancing

Table 5: Specification of the Refinancing Submodel

3.5 OTM Cash-out Refinancings

$$\begin{aligned} Cout = & Cout_Base \times Cout_WALA \times Cout_Incentive \times Cout_CLTV \\ & \times Cout_FICO \times Cout_refi_elig_pct \times Cout_HPA2yr \times Cout_Seasonals \end{aligned} \quad (8)$$

Effect	Function of	Comments
Cout_Base	NA	Baseline Constant
Cout_WALA	WALA	Seasoning
Cout_Incentive	Incentive	Incentive
Cout_CLTV	CLTV	Current LTV
Cout_FICO	FICO	Credit Score
Cout_refi_elig_pct	%RefiEligible	Eligible to refinance
Cout_HPA2yr	HPA	House price appreciation momentum
Cout_Seasonals	Seasonals	Seasonal factors for turnover

Table 6: Specification of the Cash-out Submodel

4 Model Estimation

Estimating the submodels **Curt**, **Dflt**, **Cout**, **Turn**, and **Refi** is not a straightforward task because we only observe the final prepayment rate resulting from the combination of all these sources and cannot necessarily identify the prepayment rate resulting from any particular source. The traditional workaround is to construct the model through an iterative process in which the submodels were estimated one at a time, with the residual of observed prepayment rates relative to the cumulative forecast produced by the submodels estimated so far being used to estimate the next submodel. Concretely, to begin with, the Curtailment submodel is calibrated by isolating loan-level curtailments. Similarly, we can estimate the Default submodel by isolating delinquency-related buyouts in MBS pools. Next, the Turnover submodel is estimated by differencing actual prepayment rates (on the appropriate out-of-the-money prepayment data set) with curtailment and default rates projected by the model, and then modeling this residual through explanatory variables that affect mobility rates. A similar strategy is used for estimating the Refinancing submodel by differencing overall actual prepayment rates (without any restrictions) with the curtailment, default and turnover rates projected by the relevant submodels, and then modeling the residual as the Refinancing rate. The iterative strategy concludes by estimating the Cash-out submodel.

4.1 Curtailments and Full Payoffs

Curtailment rates are directly estimated from loan-level (rather than pool-level) data recently released by Freddie Mac. The loan level curtailment calculation is as follows. For each loan that is not prepaid, we can compute its curtailment rate as a function of loan age by comparing its scheduled balance to the actual balance. This gives us the curtailment function. We calibrate a Full Payoff function by looking at prepayments on very seasoned (Weighted-average maturity > 20 years) pools. The Curtailment and Full Payoff functions are then combined into a single curve and modeled as a function of age.

Curtailments also appear to show some dependence on FICO and Current LTV. Right now, this impact is not modeled.

4.2 Defaults

Freddie Mac now releases monthly pool-level data on the unpaid principal balance (UPB) of severely delinquent mortgages that have been purchased out of a pool. This data is used to directly estimate the Default submodel.

4.3 Turnover

Starting with pool-level prepayment data on the entire universe of Fannie Mae fixed-rate pools, we partition our data set to isolate the pools that are at least 30bps out-of-the-money over the period of observation. This more or less ensures that any prepayments we are seeing correspond to either Turnover, Curtailment, OTM Cash-out Refinancings or Default. We look at the residual of these prepayments versus the cumulative projection produced by the Curtailment and Default submodels to calibrate the Turnover submodel. The estimates for the Turnover submodel obtained on this dataset are unbiased with respect to interest rates because both economic theory and empirical evidence show that interest rates affect mobility principally in out-of-the-money situations.⁴

⁴See the detailed discussion in [Cello16].

4.4 Refinancing

We now consider total prepayments on pools (without any restrictions) and look at the residual of these prepayments versus the forecast produced by summing over the Curtailment, Default, and Turnover submodels. This residual is used to calibrate the Refinancing submodel. The calibration process is an iterative process that consists of two stages (listed below).

In the first stage, we simultaneously fit the “low” Media Effect S-curve, “high” Media Effect S-curve, and the Media Effect spline.⁵ This is a 3-step process that continues to loop until the RMSE cannot be lowered any further through parameter updates.

1. Estimate the “low” Media Effect S-curve by fitting to the data set with $\text{Refinanceable\%} < 30$ (in the first iteration, assume $\text{ME} = 0$ in this region);
2. Estimate the “high” Media Effect S-curve by fitting to the data set with $\text{Refinanceable\%} > 84$ (in the first iteration, assume $\text{ME} = 1$ in this region); and,
3. Fit the Media Effect multiplier as a function of $\text{Refinanceable\%}$ on the full data set using the “low” and “high” ME effect S-curves produced from the first two stages.

Once the “low” Media Effect S-curve, “high” Media Effect S-curve, and the Media Effect spline are locked in, we estimate the other spline functions in the second stage.

4.5 OTM Cashout Refinancing

The OTM cash-out submodel is challenging to fit because cash-out prepayments can easily be confounded with both turnover and rate-term refinancings depending on the incentive level. Once the turnover and refinancing submodels are calibrated the hope is that in OTM and ATM situations, these two components capture the majority of voluntary prepayments that show a dependence on home equity growth and interest rates. Still, there are specific moments in history corresponding to very high rates of house price appreciation where the baseline speed can be as much as 2-4% CPR higher than what historical levels of turnover activity would suggest. We attempt to identify these episodes as the crucial data sets on which to fit the cash-out model.

4.6 Estimation Process

Since all the submodels are more or less estimated using a similar process, without loss of generality we’ll focus on just one. For a given submodel, we have the actual prepayments denoted by smm_i and the submodel is of the form:

$$\text{smm}_i = \alpha \prod_{j=1}^p f_j(X_{ij}) + \epsilon_i \quad (9)$$

where the X_j ’s represent variables (incentive, loan size, WALA, ...), ϵ_i is an error term with mean 0, and the f_j ’s represent unspecified smooth functions of a single X_j . α is the baseline constant and we require:

$$\alpha = 1/N \sum_{i=1}^N \text{smm}_i \quad (10)$$

The fitting algorithm is then:

⁵The “low” and “high” Media Effect thresholds noted above are determined based on visual inspection of the dependence of prepayment rates on $\text{\%Refinanceable}$.

1. **Initialize** $\hat{\alpha} = \frac{1}{N} \sum_{i=1}^N \text{smm}_i$, $\hat{f}_j \equiv 1, \forall j$
2. **Do** until $\hat{f}_j$ converge. **For** each predictor j :

$$\hat{f}_j \rightarrow \text{Smooth} \left[\frac{\text{smm}_i}{\hat{\alpha} \prod_{k \neq j} \hat{f}_k(x_{ik})} \right]$$

where **Smooth** is a smoothing operator and is typically chosen to be a Hermite spline smoother in our situation.

4.7 Issues with Estimation Process

We now discuss some issues with this approach and how we address them.

- Simple averages can be misleading in this context with a lot of clustering in the data along certain dimensions. Consequently, in practice, each smm_i observation is weighted by its balance contribution; we do not compute simple averages. Also, the algorithm itself is applied to an aggregation of the repline data observations.
- Given the flexibility in the f_j 's, we typically do not have a unique solution and the **Smooth** estimates of the f_j 's can have very non-intuitive shapes. We apply constraints to the f_j 's (monotone increasing or decreasing, linearity where appropriate) and use expert judgment to set limits for their asymptotic behavior. We also carefully screen and weight data points.
- The results depend on the order we process variables. We use expert judgment to decide which variables are more important than others and tweak the convergence algorithm so that we sequentially iterate from one f_j to another after each smoothing operating so that the fitting process “balances” the contributions of the various variables.

The upshot is that currently this estimation technique still requires a fair amount of supervision and hand-holding.

5 Appendix. Loan-level Data Coverage

Both Fannie Mae and Freddie Mac provide LL data on pools but the historical coverage of this data is dramatically different along two dimensions – the origination dates of the loans and the period over which LL history is provided:

- Freddie Mac only provides LL information for loans originated on or after September 2006 and all LL information is available on a monthly basis from this date.
- Fannie Mae provides LL information for loans originated all the way back to 1994 but this information is only available on a monthly basis starting February 2014.
 - FNMA provided a one-time LL snapshot of the entire MBS pool universe on January 2013.
 - Starting February 2014, monthly LL distributions are available on all pools that were originated in 2013 or later.
 - For pools that were originated in 2013 or later we can use the one-time snapshot to estimate the monthly LL distribution. Essentially, for any given pool, the LL distribution as of January 2013 is amortized according to the evolution of the pool factor in order to obtain a picture of the distribution for subsequent dates.

There is also a separate coverage issue in that there are partial gaps in LL coverage for pools around their origination dates – LL data can be missing for 2-3 months at a time and then start being filled in. The statistics here are (as of March 2017):

- 56% of Freddie Mac pools and 19% of Fannie Mae pools have no partial gaps.
- 40% of Freddie Mac pools and 2% of Fannie Mae pools have 1-3 month gaps.
- 3% of Freddie Mac pools and 79% of Fannie Mae pools have coverage gaps greater than 3 months.

(The vastly greater Fannie Mae gaps result from the much larger universe of Fannie Mae pools with at least some LL data).

For Ginnie Mae:

- Loan level data starts from 2013-08: it includes LL data on all outstanding pools.
- Pool-level delinquency information: starts from 2005-11 (both GNSF and G2SF). Delinquency information includes 30-day, 60-day, 90+, and Foreclosed.
- Pool-level default information (this is equal to the amount of repurchased delinquent loans): starts from 2005-11
- Loan-level delinquency information by program type (FHA/VA/RD/PIH): starts from 2013-08

6 Appendix. Definitions of Auxiliary Variables

6.1 Burnout

We accumulate the *absolute* incentive subject to a floor of 0 and a cap of 200bps:

$$I_T = \sum_{t=0}^T \min(\text{positive_incentive}_t, 200) \quad (11)$$

To account for the dependence of burnout on loan size we need to design a new burnout multiplier, which is specified as:

$$\text{burnout}_t = B(I_t, L_t) = 1 - [(1 - F(I_t)) \times G(L_t)] \quad (12)$$

Here I_t is the cumulative positive incentive at time t (subject to the same cap as above) and L_t is the weighted-average current loan size. The intuition is that burnout should depend on loan size – loans with lower loan sizes burnout slowly while those with larger loan sizes burnout quickly. F is a decreasing function with $F(0) = 1$. G is an increasing function of L_t . Note that $B(0, L_t) = 1$ and that $B(I_t, L_t) = F(I_t)$ if $G(L_t) = 1$ for all L_t . Setting $G(L_t) = 1$ allows us to model burnout without taking loan size into account.

6.2 Current LTV

We update the amortized OLTV with an estimate of the house price increase since origination. This estimate uses the quarterly purchase-only FHFA HPI.

6.3 HPA2yr

HPA Momentum, **HPA2Yr**, is used as an explanatory variable in our prepayment model as another gauge of mortgage credit availability and general housing market health (in addition to MBA’s Mortgage Credit Availability Index).

HPA2yr at time t is defined as the ratio of the House Price Index (HPI) at t and at $t - 24$:

$$\text{HPA2Yr}_t = \frac{HPI_t}{HPI_{t-24}} \quad (13)$$

6.4 Incentive

Given a conventional loan, we can define its refinancing incentive at time t as:⁶

$$\begin{aligned} \text{incentive} &= \max(\text{ConventionalIncentive}_t, \text{FHAIncentive}_t) \\ &= \max(I_t^c, I_t^g) \end{aligned} \quad (14)$$

If a loan is eligible to refinance, there is a clear-cut method for calculating its incentive – calculate the individual incentives associated with refinancing into the different programs available and then take the maximum.

$$I_t^c = \text{OrigNoteRate} + \text{OrigPMI} - (\text{PMMS}_t + \text{LLPA}_t/4 + \text{PMI}_t) \quad (15)$$

$$I_t^g = \text{OrigNoteRate} + \text{OrigPMI} - (\text{FHA Mtg Rate}_t + \text{Upfront MIP}_t/12 + \text{AnnualMIP}_t) \quad (16)$$

The core issue with this definition is that loans that are not eligible to refinance into a program do not have a clear-cut incentive with respect to that program. One workaround is to ignore programs that we do not have a well-defined incentive with respect to so that the maximum in (14) is only taken with respect to qualifying incentives. But, what happens when a loan is not eligible to refinance into any program?

If a loan is not RE with respect to any existing mortgage program than the refinance incentive is undefined. We work around this issue by introducing the notion of an **unrestricted eligibility incentive** calculation – for refinancing into a conventional program, assume that the loan has an LLPA given by the maximum value of the LLPA/PMI matrix along a particular row. For refinancing into an FHA mortgage, take the maximum MIP value along a row.

The incentive is lagged by 1.5 months and LLPA calculations are smoothed using a bilinear interpolation of the LLPA matrix.

6.5 Media Effect

The **Media Effect** measures how low prevailing mortgage rates are relative to past history. A proxy measure for the magnitude of this effect at a given point in time is calculating the “in-the-moneyness” of the universe of existing mortgages. Here, “moneyness” is calculated as the fraction of the mortgage universe that is at least 50bps in-the-money.

6.6 monthsSince

The variable **monthsSince** is constructed for the default data set. Fix an **asofdate** and a pool that is 100% current for that date. We create a time series for that pool that corresponds to its default rates for

⁶We do not incorporate the notion of HARP refinanceability since HARP-related outstanding balances have shrunk considerably and the program is intended to be sunset by the end of 2018.

every `asofdate` succeeding that date. `monthsSince` increments by one for each such date. Now move to `asofdate + 1` month. If the pool is still 100% current, repeat the process above to come up with another time series of default rates and `monthsSince` values. Thus, each pool creates multiple time-series. The rationale for this setup is to get a sense for the baseline default rates produced by pools that we know to initially be 100% current. A similar `monthsSince` calculation applies for pools that are partially delinquent. `monthsSince` is therefore setting up a baseline default rate for a pool based on its last recorded delinquency status.

`monthsSince = 0` when we look at CDR values on pools for a given historical factor date. This leads to two default files that we use to visualize the tracking. The standard “`asofdate`” file simply aggregates CDR values on pools across historical factor dates (As noted above, we do not use LL data in the conventional case because the `loanRemovalReason` for conventional loans is not populated as opposed to Ginnie Maes where it is). A second file aggregates across all `monthsSince` values (the first file has `monthsSince = 0` by definition).

6.7 Mortgage Credit Availability

Mortgage Credit Availability (MCA) can be defined as the supply of mortgage credit available for a reference household, after controlling for economic and demographic factors. In principle, the control is necessary so that the index isolates the liberalization or contraction of credit in absolute terms and does not simply capture the stylized economic facts that the provision of credit varies over the business cycle and that credit supply also depends on borrower demographics. Still, it may beg the question to disentangle the role of economic factors from credit supply given the significant role that this supply itself plays in the macroeconomy. In other words, a mortgage credit boom may in and of itself stimulate the economy, cf. the “housing bubble years” of 2001-2006. Similarly, the pricking of a credit bubble may throw the economy into a recession/depression, as in the case of the Great Depression (1929-1939) or the Great Recession (2007-2009)[Mian15].

A **Mortgage Credit Availability Index (MCAI)** is designed to measure the supply of mortgage credit in a systematic way over time. The motivation in doing so stems from housing and general economic policy considerations as housing has time and time again shown itself to play a crucial role in driving the business cycle. Regulators tracking such an index would presumably attempt to apply macroprudential restrictions to cool down an overheated mortgage market or to stimulate a moribund one. From Cello’s vantage point, the main interest is in correlating changes in MCAI to changes in refinancing efficiency. It is clear that there is a strong positive correlation between MCA and the ability to refinance (and probably, to a much lesser extent, to turnover). An increase in MCA either through relaxing underwriting standards or by introducing new mortgage products that allow existing mortgage holders to refinance with less friction than in the past. Tracking this type of structural change in refinancing efficiency is critical in terms of forecasting prepayment rates, particularly when interest rates have held relatively steady for a prolonged period of time. The point being that increases/decreases in MCA may not be visible for a while in terms of observable refinancing rates because most borrowers do not view existing refinancing opportunities as attractive, and then all of sudden express themselves once a significant fraction of mortgagors is able to refinance.

The MBA’s MCAI is constructed using underwriting criteria from more than 95 lenders and investors.⁷ The MCAI is based on changes in the underwriting guidelines for institutional investors who purchase loans through the broker and/or correspondent channels. Higher index values signal that credit is more available, while lower index values indicate that mortgage credit standards are tighter. In March 2012, the MBA set the MCAI to 100.00. After that month, the MCAI either moves up or down according to the change in credit availability conditions during that month relative to the base monthly. An index of 110, for example, means there has been a 10-percent increase in availability of credit since the reference period (March 2012); similarly, an index of 90 means a 10-percent decrease.

The data for the index calculation is provided by AllRegs Market Clarity a platform that provides underwriting guidelines from more than 85 investors into one place. The main dimensions of mortgage credit risk taken into

⁷See <https://www.mba.org/news-research-and-resources/research-and-economics/single-family-research/mortgage-credit-availability-index>.

consideration are loan purpose (Purchase, Rate / Term Refinance, Cash-Out Refinance), amortization type (Fixed, ARM, Balloon), property type (primary residence, second home, investor property), loan term (years to maturity), LTV, credit score, documentation type (full, stated, no-doc) and payment type (interest-only vs. non). MBA economists assign a risk weight to each credit qualification metric so that loan programs that have had historically higher delinquency rates are given greater weight in the index. For example, loan programs with low minimum credit scores are given a greater risk weight than those with higher minimum credit scores. Some dimensions of credit qualification are examined together (layered risks are considered).

6.8 pct_HARPed

Loans that have been refinanced through the Home Affordable Refinance Program (HARP) generally prepay slower than other refinance loans that have undergone full underwriting. As a streamline refinance program, HARP generally made only minimal requirements of borrowers, and such borrowers may subsequently have trouble qualifying for regular refinances. Thus, it is important to identify the percentage of HARPed borrowers in any given repline.

6.9 Refinance Eligibility

The **Refinancing Eligibility (RE)** of a mortgage loan is a simple 0-1 indicator that captures whether a loan is refinaneable (RE=1) based on prevailing underwriting standards. The notion of RE is independent of incentive and serves as a check of whether a borrower and loan's characteristics falls within the boundaries of the current underwriting matrix. Some of the factors considered in this determination include FICO, OLTV, Property Type, Occupancy Status, Debt-to-Income Ratios etc. Since underwriting standards evolve over time, RE needs to be captured through a time series.

A conventional loan is considered to be **refinance eligible** if it is refinaneable through *at least one* of the following programs:

1. **Standard Conventional.** This assesses whether a loan is eligible to refinance into a conventional program (i.e., into a loan guaranteed by Fannie Mae or Freddie Mac) based on prevailing underwriting standards. To ascertain this, we look at the FICO and LTV bounds associated with different Loan Purposes, Property Types, Occupancy Statuses etc.⁸
2. **HARP.** This determines whether a loan qualifies for a HARP refinance. The basic HARP eligibility criteria are:
 - The loan must be owned or guaranteed by Fannie Mae or Freddie Mac;
 - The loan must have been originated on or before May 31, 2009;
 - The current LTV must be greater than 80%.⁹ There is no LTV ceiling;
 - Borrowers must be current on their mortgage payments at the time of the refinance; and,
 - Payment history – a borrower is allowed one late payment in the past 12 months, as long as it did not occur in the 6 months prior to the refinance.

In practice, since loan-level delinquency data is not available for the loans corresponding to GSE pools, Cello's test for HARP-eligibility will simply check for the date of loan origination. The HARP program is scheduled to be sunset on December 31st, 2018 so the notion of HARP-eligibility is distinctly time-bound: it only applies from June 2009 to December 2018. Note that government loans pooled in GNMA MBSs are not eligible for HARP.

⁸This information is embedded in Cello's time-series of LLPA matrices.

⁹In practice, LTVs below 80% also use the HARP program as long as they meet the other requirements. The main difference for these loans appears to be the fact that such refinances are not directly solicited by lenders.

3. **FHA.**¹⁰ This identifies whether the conventional loan can refinance into the FHA program.¹¹ The official government-imposed minimum credit score for an FHA home loan is 500. However, such loans have to put 10% down. To qualify for an LTV of 97.5%, a score of 580 or higher is needed. Mortgage lenders can set their own minimum credit-score requirements for FHA loans and they are usually higher than official cutoff. Most lenders will want to see a score of 600 or higher. Finally, the property has to be owner-occupied.¹²

Based on the above description of RE, for each loan, we can associate a time-series of 3-tuples of Boolean values corresponding to its refinancability under the individual programs. The time-series notion captures the fact that the refinancability of an individual loan evolves as its collateral characteristics change (we can only estimate CLTV changes based on the current mortgage data set shared by the GSEs) and underwriting guidelines change (LTV/FICO cap and/or floors are changed etc.). A loan has RE=1 if it meets the underwriting criteria for *at least one* of the 3 options above.

Clearly, there is a natural way to scale this calculation to any aggregation (whether replines or pools) except that now the RE indicator is a “continuous” variable between $[0, 1]$ (corresponding to the % of the repline or pool that is RE) as opposed to the discrete indicator for loans.

6.10 Tax Multiplier for Housing Turnover

The 1995-1997 time period corresponds to a structural shift in the housing market in May 1997 with the passage of capital gains taxation reform. Academic studies show that there was a significant positive impact on mobility after the law was implemented with a couple of estimates suggesting a 20-25% increase in the rate of home sales compared to the baseline levels prevailing prior to the passage of the law (for example, see [Shan11]). This leads us to introduce an origination date based multiplier (which we name as **tax**) which is calibrated to 0.77 prior to May 1997 and 1.0 after that.

7 Appendix. Default Data Set

There are potentially two sources of data for estimating a default model – the standard prepayment loan- and pool-level data set updated every factor date by the Agencies and the loan-level **Credit Data Set** released by the Agencies to help in estimating models for mortgage credit. The standard prepayment data set is the one we rely on to estimate the model since it is closest to the population of loans/pools we are attempting to predict prepayments for. However, the Credit Data Set contains more granular data and covers a greater span of history.

7.1 Default Data in the Prepayment Data Set

- Delinquency/Default information only available for Freddie pools
- $\approx 352K$ pools with data as far back as 2011
- 30 Day, 60 Day, 90 Day, 120 Day and Repurchased balances are available. These are exclusive, not cumulative percentages. The 120 Day Balance is usually 0 since the Agencies will removed 120 Day delinquent loans from the pool but there are exceptions¹³

¹⁰We do not consider VA-, RHS, or PIH-eligibility for conventional loans because there are special requirements for qualifying for these programs which we are unable to check based on the current data set provided to us by the GSEs.

¹¹Note that we are only talking about *eligibility* here; it is of course possible that there may be greater incentive for a conventional loan to refinance into an FHA loan than another conventional one.

¹²The FHA does not allow a non-FHA investor or 2nd home to refinance into an FHA mortgage.

¹³See Appendix 8.

- The `pct_dq%` is the sum of the 30 Day, 60 Day, 90 Day and 120 Day balances.
- No Delinquency/Default information is available at the loan level

7.2 Default Data in the Credit Data Set

- Delinquency/Default information available for Fannie and Freddie loans
- $\approx$ 40 million loans with data as far back as 1999
- Loan Age up to 180 months
- Separate loan-level indicators for voluntary and involuntary (Default) prepayment
- Excellent coverage for FICO, DTI, CLTV (MSA level), Occupancy, etc.
- The default definition in this data set corresponds to 180 day delinquent loans being removed from the data and a Zero Balance Code being reported. This criteria needs to be modified to match with the pool data. The first time a loan reaches 120 days delinquent it is marked as having defaulted.

8 Appendix. GSE Buyouts in 2010

One of the biggest drivers of prepayments in the first half of 2010 was involuntary prepayments occurring due to the buyout of 120+ delinquent loans by the GSEs. Freddie Mac bought out all the 120+ delinquent loans backing their MBS in February and then continued to buyout any incremental 120+ delinquencies on a monthly basis. Fannie Mae executed one-time buyouts from March to June and, similar to Freddie Mac, continued to buyout the incremental 120+ delinquencies from their pools every month.

The GSEs said it made economic sense for them to do this because they were advancing payments to investors at above-market rates. Given that these companies were (and are) backed by the U.S. Treasury at this point in time, their cost of funding was a lot lower. Given this low cost, it made sense to buyout these delinquent mortgages as long as there was sufficient capital and the advancing rate was higher than the cost of funding.

To calculate “buyout CPRs” at the pool level, use the prior months 120+ day delinquency rate to figure out next months buyout. For data prior to Jan 2011 (Freddie Mae started releasing pool-level data as of this date), the GSEs used to publish the 120+ day delinquency percentages by coupon and vintages in a PDF document on their website. The magnitude of this activity varies heavily by origination year, coupon and calendar time (the relevant month in 2010) and generally appears to average between 5-10% CPR.

Prepayment models will generally completely miss this surge in prepayments because it essentially corresponded to an exogenous shock to prepayment rates as the result of a policy change with the GSEs essentially flushing out as much of their 120+ delinquent pipeline as possible.

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